

MOHAMED A. MOHAMED

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SUMMARY

Infrastructure engineer specializing in AWS, Linux, Kubernetes/EKS, Infrastructure as Code, CI/CD, and secure cloud platforms. Hands-on experience building and troubleshooting cloud-native platforms using Terraform, GitOps, automation, observability, and production-style Kubernetes infrastructure.

SKILLS

Cloud & Infrastructure: AWS, EKS, EC2, VPC, IAM, S3, ECR, RDS, Terraform, CloudFormation

Containers & Platforms: Linux, Docker, Kubernetes, Helm, Kustomize, Argo CD, Backstage

Source Control & CI/CD: Git, GitHub, GitHub Actions, GitOps, CI/CD pipelines, OIDC

Monitoring & Observability: Prometheus, Grafana, metrics, logging, health probes, incident troubleshooting

Web & Automation: HTTP/HTTPS, REST APIs, FastAPI, Bash, Python, Boto3, Linux CLI

EXPERIENCE

IT SYSTEMS SPECIALIST | Galoof & Associates | Boston, MA | January 2021 - Present

- Supported business-critical infrastructure across 11 workstations, 3 servers, networking, identity, backups, applications, and cloud-managed services for a nine-user office.
- Troubleshot operating systems, access, applications, networking, connectivity, and server-dependent services using system logs, diagnostic tools, and structured troubleshooting
- Automated recurring maintenance, monitoring, configuration, backup, and compliance tasks to reduce manual administration and improve operational consistency.
- Supported Microsoft 365, Outlook, Entra ID, Intune, Identity services, and operational recovery procedures during high-volume tax seasons.

PROJECTS

FORGEPath | Secure Platform Engineering Paved Path | github.com/SecureCloudOps/ForgePath

- Built a Backstage-based paved path that scaffolds production-ready FastAPI services and delivers digest-pinned container workloads to Kubernetes through Helm, GitOps, and Argo CD.
- Engineered fail-closed validation gates spanning automated tests, SAST, secret and vulnerability scanning, OPA policy checks, SPDX SBOM generation, Cosign signing, and digest-bound artifact promotion.
- Hardened Kubernetes delivery with Kyverno admission controls, least-privilege RBAC, default-deny networking, health probes, structured JSON logging, and Argo CD self-healing.

KUBEBackup OPERATOR | Kubernetes Database Backup Operator | github.com/SecureCloudOps/database-backup-operator

- Engineered a production-grade Kubernetes operator in Go using Kubebuilder and controller-runtime, automating declarative PostgreSQL backup and restore workflows across MinIO and Amazon S3.
- Improved recovery reliability and workload security through SHA-256 restore verification, Prometheus observability, least-privilege RBAC, default-deny NetworkPolicies, and hardened non-root containers.
- Validated backup durability and exact data recovery with 21 integration tests and 3 Kind E2E scenarios, achieving 68.5% controller coverage; automated CI/CD security scanning, SBOM generation, build provenance, image signing, and digest-pinned releases.

LLM RAG INFRASTRUCTURE PLATFORM | github.com/SecureCloudOps/llm-rag-infra-platform

- Architected a production-style RAG platform using FastAPI, Qdrant, vLLM, Docker, Kubernetes, and AWS EKS, exposing APIs for document ingestion, semantic retrieval, and LLM-generated responses.
- Automated AWS infrastructure provisioning with modular Terraform across VPC networking, IAM, ECR, persistent storage, and EKS, enabling repeatable GitOps deployments through Argo CD.
- Established CI quality gates for application tests, container builds, Kubernetes manifests, Terraform, and security scans; implemented Prometheus and Grafana observability for request rates, latency, errors, application health, and Kubernetes workloads.

CERTIFICATIONS

AWS Certified SysOps Administrator – Associate | 2025

Certified Kubernetes Administrator (CKA) | 2025

CompTIA Security+ | 2025

AWS Certified Solutions Architect – Associate | 2024
